EXPERIMENT NO: 12

Experiment Title: Build an interactive real-time web app using JavaScript and jQuery with DOM, events, storage, and AJAX integration.

Software/Tools Required:

- Any text editor / VS Code
- Web Browser (Chrome/Edge) with Developer Console

Theory:

JavaScript and jQuery: JavaScript is a scripting language used to create dynamic and interactive web applications. jQuery is a JavaScript library that simplifies DOM manipulation, event handling, animations, and AJAX operations.

DOM Manipulation: The Document Object Model (DOM) represents an HTML document as a tree of elements. JavaScript and jQuery can access, modify, add, or remove HTML elements dynamically without reloading the webpage. In this experiment, DOM manipulation is used to display and update a list of items.

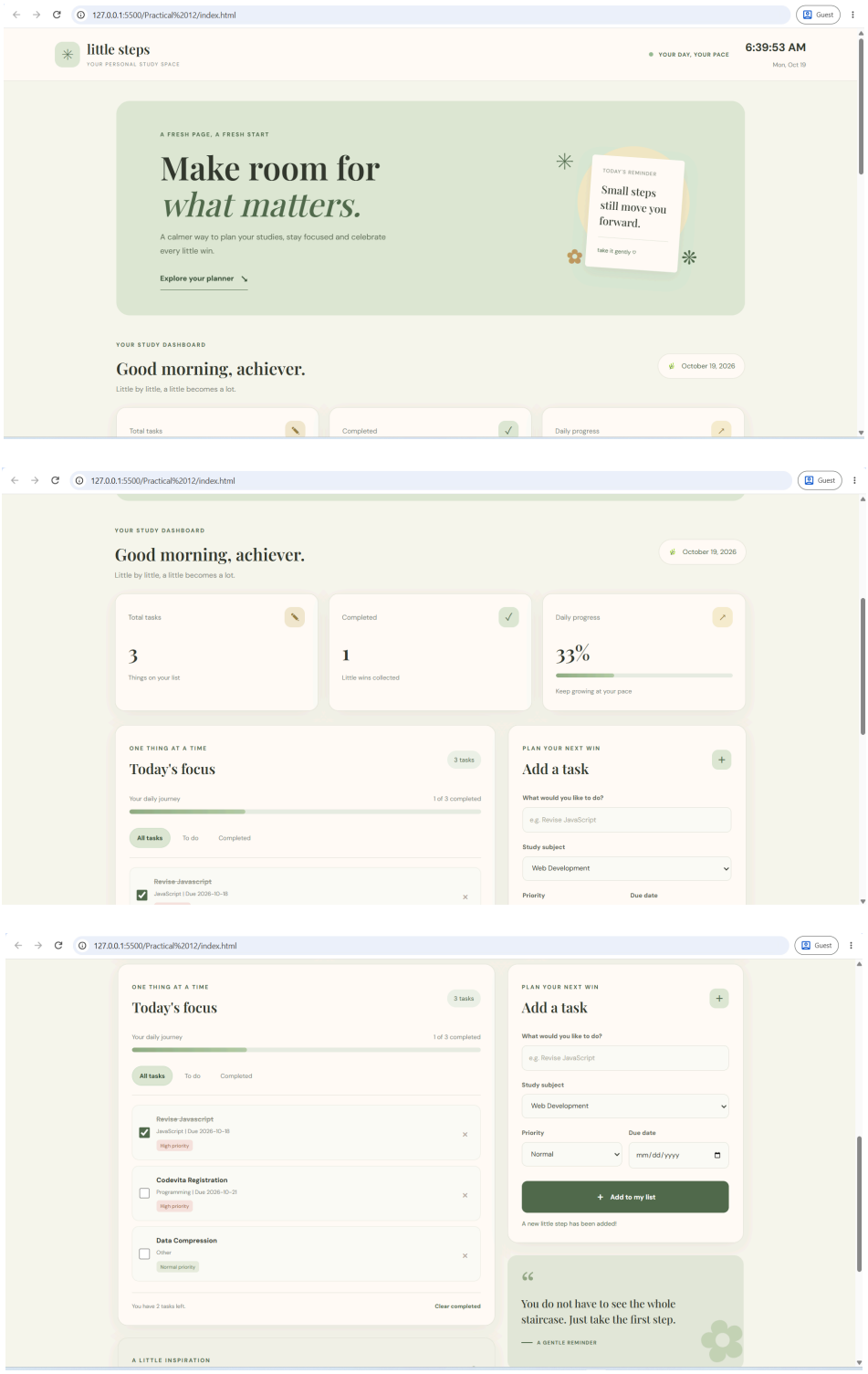
Event Handling: Events allow a web application to respond to user actions such as clicks, typing, and form submissions. jQuery provides methods such as on() to handle events. In this experiment, event handling is used to add and delete items interactively.

Web Storage (localStorage): Web Storage allows data to be stored in the user's browser. The localStorage object stores data as key-value pairs and retains it even after the browser is refreshed or closed. In this experiment, localStorage is used to save the list of items. The JSON.stringify() and JSON.parse() methods convert data between JavaScript objects and JSON strings.

AJAX Integration: AJAX (Asynchronous JavaScript and XML) allows a web application to exchange data with a server without reloading the entire webpage. jQuery provides the $.ajax() method to send asynchronous requests and process server responses. In this experiment, AJAX is used to fetch task information from an external API and display it dynamically.

Real-Time Interactive Web Application: An interactive web application combines DOM manipulation, event handling, browser storage, and asynchronous data retrieval to provide a responsive user experience. In this experiment, users can add and delete items, preserve them using browser storage, and fetch external data using AJAX.

OUTPUT:



RESULT:

Thus, an interactive real-time web application was successfully implemented using JavaScript and jQuery with DOM manipulation, event handling, browser storage using localStorage, and AJAX integration for asynchronous data retrieval.